Statistics/Data analysis

```
      name:  <unnamed>
       log:  E:\Documents\STATA\assignment 2_Number one baby outcomes.smcl
  log type:  smcl
 opened on:  14 May 2026, 09:58:29

1 .
2 . *import data
3 . import excel "E:\Documents\STATA\assignment 2_Number one_2025_2026_revised.xls", she
  > et("Sheet1") firstrow
  (15 vars, 669 obs)

4 .
5 . *inspect data
6 . bro

7 .
8 .
9 . *RENAME VARIABLES
10.
11. rename Column3  platelet

12. rename Column4  race

13. rename Column5  birthwt

14. rename Column6  gestage

15. rename Column7  placedev

16. rename Column8  multpreg

17. rename Column9  betamet

18. rename Column10 tocolys

19. rename Column11 modedev

20. rename Column12 apgar1

21. rename Column13 ventil

22. rename Column14 pneumo

23. rename Column15 pda

24. rename Column16 sexbaby

25. rename Column17 died

26.
27.  *LABEL VARIABLES
28.
29. label variable platelet  "Platelet count"

30. label variable race      "Race of mother (1=white,2=black,3=native American,4=orient
  > al)"

31. label variable birthwt   "Birth weight of baby (grams)"
```

```
32. label variable gestage   "Gestation age"
33. label variable placedev  "Place of delivery (1=this hospital,2=elsewhere)"
34. label variable multpreg  "Multiple pregnancy (1=yes,0=no)"
35. label variable betamet   "Mother treated with betamethasone (1=yes,0=no)"
36. label variable tocolys   "Tocolysis (1=yes,0=no)"
37. label variable modedev   "Mode of delivery (1=abdominal,2=vaginal)"
38. label variable apgar1    "Apgar score at 1 minute"
39. label variable ventil    "Assisted ventilation used (1=yes,0=no)"
40. label variable pneumo    "Pneumothorax occurred (1=yes,0=no)"
41. label variable pda       "Patent ductus arteriosus occurred (1=yes,0=no)"
42. label variable sexbaby   "Sex of baby (1=female,2=male)"
43. label variable died      "Baby died (1=yes,0=no)"
44.
45. *DATA INSPECTION
46.
47. bro
48.
49. * Continuous variables
50. summarize platelet birthwt gestage apgar1
```

Variable	Obs	Mean	Std. dev.	Min	Max
platelet	**669**	**199.6565**	**80.95436**	**16**	**571**
birthwt	**669**	**1093.886**	**265.2247**	**400**	**1580**
gestage	**669**	**28.87668**	**2.542624**	**22**	**40**
apgar1	**669**	**4.908819**	**2.624002**	**0**	**9**

```
51.
52. * Categorical variables
53. tab race
```

Race of mother (1=white,2=black,3=native American,4=oriental)	Freq.	Percent	Cum.
1	**260**	**38.86**	**38.86**
2	**372**	**55.61**	**94.47**
3	**26**	**3.89**	**98.36**
4	**11**	**1.64**	**100.00**
Total	**669**	**100.00**	

```
54. tab placedev
```

Place of delivery (1=this hospital,2=elsewhere)	Freq.	Percent	Cum.
1	**546**	**81.61**	**81.61**
2	**123**	**18.39**	**100.00**
Total	**669**	**100.00**	

55. tab multpreg

Multiple pregnancy (1=yes,0=no)	Freq.	Percent	Cum.
0	**531**	**79.37**	**79.37**
1	**138**	**20.63**	**100.00**
Total	**669**	**100.00**	

56. tab betamet

Mother treated with betamethasone (1=yes,0=no)	Freq.	Percent	Cum.
0	**373**	**55.75**	**55.75**
1	**296**	**44.25**	**100.00**
Total	**669**	**100.00**	

57. tab tocolys

Tocolysis (1=yes,0=no)	Freq.	Percent	Cum.
0	**508**	**75.93**	**75.93**
1	**161**	**24.07**	**100.00**
Total	**669**	**100.00**	

58. tab modedev

Mode of delivery (1=abdominal,2=vaginal)	Freq.	Percent	Cum.
1	**323**	**48.28**	**48.28**
2	**346**	**51.72**	**100.00**
Total	**669**	**100.00**	

59. tab ventil

Assisted ventilation used (1=yes,0=no)	Freq.	Percent	Cum.
0	**284**	**42.45**	**42.45**
1	**385**	**57.55**	**100.00**
Total	**669**	**100.00**	

```
60. tab pneumo
```

Pneumothorax occurred (1=yes,0=no)	Freq.	Percent	Cum.
0	**537**	**80.27**	**80.27**
1	**132**	**19.73**	**100.00**
Total	**669**	**100.00**	

```
61. tab pda
```

Patent ductus arteriosus occurred (1=yes,0=no)	Freq.	Percent	Cum.
0	**527**	**78.77**	**78.77**
1	**142**	**21.23**	**100.00**
Total	**669**	**100.00**	

```
62. tab sexbaby
```

Sex of baby (1=female,2=male)	Freq.	Percent	Cum.
1	**331**	**49.48**	**49.48**
2	**338**	**50.52**	**100.00**
Total	**669**	**100.00**	

```
63. tab died
```

Baby died (1=yes,0=no)	Freq.	Percent	Cum.
0	**527**	**78.77**	**78.77**
1	**142**	**21.23**	**100.00**
Total	**669**	**100.00**	

```
64.
65. * Missing values
66. misstable summarize platelet race birthwt gestage placedev multpreg betamet tocolys
  > modedev apgar1 ventil pneumo pda sexbaby died
  (variables nonmissing or string)

67.
68. * The outcome variable (died) has no missing values - all 669 observations are avail
  > able for the outcome
69.
70. * PRE-ANALYSIS DECISION I: WHICH RACE CODING IS BEST?
71. * The question asks us to compare 4 ways of coding race.
```

```
72. * Since these 4 models are non-nested, we compare using AIC and BIC.
73. * The model with the smallest BIC is preferred.
74.
75. * Generate the 3 alternative race codings
76.
77. * Option ii: White (0) vs All other races (1)
78. gen race_white_other = 0 if race == 1
  (409 missing values generated)

79. replace race_white_other = 1 if race == 2 | race == 3 | race == 4
  (409 real changes made)

80. label variable race_white_other "Race: White(0) vs Other races(1)"

81.
82. * Option iii: White (1) vs Black (2) vs Other races (3)
83. gen race_white_black_other = 1 if race == 1
  (409 missing values generated)

84. replace race_white_black_other = 2 if race == 2
  (372 real changes made)

85. replace race_white_black_other = 3 if race == 3 | race == 4
  (37 real changes made)

86. label variable race_white_black_other "Race: White(1) vs Black(2) vs Other(3)"

87.
88. * Option iv: Black (1) vs All other races (0)
89. gen race_black_other = 0 if race == 1 | race == 3 | race == 4
  (372 missing values generated)

90. replace race_black_other = 1 if race == 2
  (372 real changes made)

91. label variable race_black_other "Race: Non-black(0) vs Black(1)"

92.
93. * Verify all three were generated correctly
94. tab race_white_other
```

Race: White(0) vs Other races(1)	Freq.	Percent	Cum.
0	**260**	**38.86**	**38.86**
1	**409**	**61.14**	**100.00**
Total	**669**	**100.00**	

```
95. tab race_white_black_other
```

Race: White(1) vs Black(2) vs Other(3)	Freq.	Percent	Cum.
1	**260**	**38.86**	**38.86**
2	**372**	**55.61**	**94.47**
3	**37**	**5.53**	**100.00**
Total	**669**	**100.00**	

```
96. tab race_black_other
```

Race: Non-black(0) vs Black(1)	Freq.	Percent	Cum.
0	**297**	**44.39**	**44.39**
1	**372**	**55.61**	**100.00**
Total	**669**	**100.00**	

```
97.
98. * Fit all 4 models using the full set of predictors - only race coding changes
99. * Option i: Race as it is (4 categories)
100 logit died i.race platelet birthwt gestage i.placedev i.multpreg i.betamet i.tocolys
  >  i.modedev apgar1 i.ventil i.pneumo i.pda i.sexbaby

  Iteration 0:   log likelihood =  -345.8274
  Iteration 1:   log likelihood =  -224.3776
  Iteration 2:   log likelihood =  -203.5184
  Iteration 3:   log likelihood =   -202.627
  Iteration 4:   log likelihood = -202.62519
  Iteration 5:   log likelihood = -202.62519

  Logistic regression                                     Number of obs =    669
                                                          LR chi2(16)   = 286.40
                                                          Prob > chi2   = 0.0000
  Log likelihood = -202.62519                             Pseudo R2     = 0.4141
```

died	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
race						
2	.5398888	.2959089	1.82	0.068	-.0400819	1.11986
3	1.531819	.5977242	2.56	0.010	.3603007	2.703336
4	1.759886	.864387	2.04	0.042	.0657186	3.454053
platelet	-.0017418	.0017536	-0.99	0.321	-.0051787	.0016951
birthwt	-.001459	.0008503	-1.72	0.086	-.0031256	.0002076
gestage	-.5210846	.1070366	-4.87	0.000	-.7308726	-.3112967
2.placedev	.2323468	.3109829	0.75	0.455	-.3771685	.8418621
1.multpreg	.4922861	.3473595	1.42	0.156	-.188526	1.173098
1.betamet	-.3393345	.2806127	-1.21	0.227	-.8893254	.2106563
1.tocolys	-.1278382	.3210421	-0.40	0.690	-.7570691	.5013928
2.modedev	-.0180172	.2797831	-0.06	0.949	-.5663819	.5303476
apgar1	-.0356818	.057366	-0.62	0.534	-.1481172	.0767535
1.ventil	.8334484	.381998	2.18	0.029	.084746	1.582151
1.pneumo	1.797991	.305966	5.88	0.000	1.198309	2.397673
1.pda	-.3273395	.2881446	-1.14	0.256	-.8920926	.2374136
2.sexbaby	-.0538757	.265134	-0.20	0.839	-.5735288	.4657775
_cons	13.74318	2.525019	5.44	0.000	8.794234	18.69213

```
101 estimates store race_4cat

102
103 * Option ii: White vs Other races
104 logit died i.race_white_other platelet birthwt gestage i.placedev i.multpreg i.betam
  > et i.tocolys i.modedev apgar1 i.ventil i.pneumo i.pda i.sexbaby

  Iteration 0:   log likelihood =  -345.8274
  Iteration 1:   log likelihood = -225.79538
  Iteration 2:   log likelihood = -205.75002
  Iteration 3:   log likelihood = -205.03588
  Iteration 4:   log likelihood = -205.03418
  Iteration 5:   log likelihood = -205.03418
```

```
Logistic regression                                     Number of obs =    669
                                                        LR chi2(14)   = 281.59
                                                        Prob > chi2   = 0.0000
Log likelihood = -205.03418                             Pseudo R2     = 0.4071
```

died	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
1.race_white_other	.6674574	.2889324	2.31	0.021	.1011603	1.233755
platelet	-.0018765	.0017351	-1.08	0.279	-.0052772	.0015242
birthwt	-.0015043	.0008374	-1.80	0.072	-.0031455	.0001369
gestage	-.5211553	.1042435	-5.00	0.000	-.7254688	-.3168418
2.placedev	.2587623	.3099849	0.83	0.404	-.348797	.8663216
1.multpreg	.4112682	.3440374	1.20	0.232	-.2630326	1.085569
1.betamet	-.2702313	.2768959	-0.98	0.329	-.8129374	.2724747
1.tocolys	-.1623952	.3172102	-0.51	0.609	-.7841158	.4593254
2.mododev	-.0363512	.2778797	-0.13	0.896	-.5809855	.508283
apgar1	-.0326145	.0570096	-0.57	0.567	-.1443511	.0791222
1.ventil	.6936039	.3660787	1.89	0.058	-.0238972	1.411105
1.pneumo	1.826412	.3078401	5.93	0.000	1.223057	2.429768
1.pda	-.3204829	.2878455	-1.11	0.266	-.8846497	.2436838
2.sexbaby	-.0524902	.2626692	-0.20	0.842	-.5673124	.4623319
_cons	13.90303	2.473213	5.62	0.000	9.055622	18.75044

```
105 estimates store race_2cat_white

106
107 * Option iii: White vs Black vs Other
108 logit died i.race_white_black_other platelet birthwt gestage i.placedev i.multpreg i
  > .betamet i.tocolys i.mododev apgar1 i.ventil i.pneumo i.pda i.sexbaby

  Iteration 0:   log likelihood =  -345.8274
  Iteration 1:   log likelihood = -224.47493
  Iteration 2:   log likelihood = -203.54187
  Iteration 3:   log likelihood = -202.65274
  Iteration 4:   log likelihood = -202.65097
  Iteration 5:   log likelihood = -202.65097

  Logistic regression                                     Number of obs =    669
                                                          LR chi2(15)   = 286.35
                                                          Prob > chi2   = 0.0000
  Log likelihood = -202.65097                             Pseudo R2     = 0.4140
```

died	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
race_white_black_other						
2	.5432139	.2957105	1.84	0.066	-.0363681	1.1227
3	1.601926	.5102698	3.14	0.002	.601816	2.6020
platelet	-.001694	.0017417	-0.97	0.331	-.0051078	.00171
birthwt	-.0014415	.0008471	-1.70	0.089	-.0031017	.00021
gestage	-.5248212	.1059072	-4.96	0.000	-.7323956	-.31724
2.placedev	.2295473	.3105838	0.74	0.460	-.3791858	.83828
1.multpreg	.4891163	.3471322	1.41	0.159	-.1912504	1.1694
1.betamet	-.343665	.2798652	-1.23	0.219	-.8921907	.20486
1.tocolys	-.1186554	.3183101	-0.37	0.709	-.7425317	.50522
2.mododev	-.0216927	.2792629	-0.08	0.938	-.5690379	.52565

apgar1	-.0347054	.0572185	-0.61	0.544	-.1468515	.07744
1.ventil	.8313328	.3816744	2.18	0.029	.0832647	1.5794
1.pneumo	1.801263	.3057134	5.89	0.000	1.202076	2.4004
1.pda	-.3230636	.2874966	-1.12	0.261	-.8865466	.24041
2.sexbaby	-.0582843	.2644548	-0.22	0.826	-.5766062	.46003
_cons	13.81834	2.505519	5.52	0.000	8.907612	18.729

```
109 estimates store race_3cat

110
111 * Option iv: Black vs Rest
112 logit died i.race_black_other platelet birthwt gestage i.placedev i.multpreg i.betam
  > et i.tocolys i.mododev apgar1 i.ventil i.pneumo i.pda i.sexbaby

  Iteration 0:   log likelihood =  -345.8274
  Iteration 1:   log likelihood = -227.64572
  Iteration 2:   log likelihood = -208.17553
  Iteration 3:   log likelihood = -207.47065
  Iteration 4:   log likelihood = -207.46869
  Iteration 5:   log likelihood = -207.46869

  Logistic regression                                     Number of obs =     669
                                                          LR chi2(14)   =  276.72
                                                          Prob > chi2   =  0.0000
  Log likelihood = -207.46869                             Pseudo R2     =  0.4001
```

died	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
1.race_black_other	.217086	.2672294	0.81	0.417	-.306674	.740846
platelet	-.0018768	.0017189	-1.09	0.275	-.0052458	.0014923
birthwt	-.0016059	.00083	-1.93	0.053	-.0032327	.0000209
gestage	-.5227005	.1032433	-5.06	0.000	-.7250535	-.3203474
2.placedev	.2423597	.3095612	0.78	0.434	-.3643692	.8490886
1.multpreg	.4263538	.3422497	1.25	0.213	-.2444433	1.097151
1.betamet	-.2372241	.2757817	-0.86	0.390	-.7777464	.3032981
1.tocolys	-.231783	.3154361	-0.73	0.462	-.8500265	.3864604
2.mododev	.0148372	.2758684	0.05	0.957	-.5258549	.5555293
apgar1	-.0450826	.0566265	-0.80	0.426	-.1560685	.0659034
1.ventil	.6482451	.3644172	1.78	0.075	-.0659994	1.36249
1.pneumo	1.717101	.3011018	5.70	0.000	1.126952	2.307249
1.pda	-.3098821	.2849495	-1.09	0.277	-.868373	.2486087
2.sexbaby	-.0166445	.2607285	-0.06	0.949	-.5276629	.4943739
_cons	14.42524	2.447611	5.89	0.000	9.628013	19.22247

```
113 estimates store race_2cat_black

114
115 * Compare all four models - smaller AIC and BIC indicate better fit
116 estimates stats race_4cat race_2cat_white race_3cat race_2cat_black

  Akaike's information criterion and Bayesian information criterion
```

Model	N	ll(null)	ll(model)	df	AIC	BIC
race_4cat	669	-345.8274	-202.6252	17	439.2504	515.8487
race_2cat_~e	669	-345.8274	-205.0342	15	440.0684	507.6551
race_3cat	669	-345.8274	-202.651	16	437.3019	509.3945
race_2cat_~k	669	-345.8274	-207.4687	15	444.9374	512.5241

```
  Note: BIC uses N = number of observations. See [R] BIC note.
```

```
117
118 * Both AIC and BIC agreed that option (ii) - White vs Other races - provided
119 * the best fit. race_white_other will be used in all subsequent analysis.
120
121 * PRE-ANALYSIS DECISION II: BIRTHWEIGHT - CONTINUOUS OR CATEGORICAL?
122 * The question asks whether birthweight is better analyzed as continuous
123 * or categorized into <1000g vs >=1000g. AIC and BIC are used to compare.
124
125 * Generate categorical birthweight variable
126 gen birthwt_cat = 1 if birthwt >= 1000
  (231 missing values generated)

127 replace birthwt_cat = 0 if birthwt < 1000
  (231 real changes made)

128 label variable birthwt_cat "Birthweight: below 1000g(0) vs 1000g and above(1)"

129
130 * Model with birthweight as continuous
131 logit died i.race_white_other platelet birthwt gestage i.placedev i.multpreg i.betam
  > et i.tocolys i.modedev apgar1 i.ventil i.pneumo i.pda i.sexbaby

  Iteration 0:   log likelihood =  -345.8274
  Iteration 1:   log likelihood = -225.79538
  Iteration 2:   log likelihood = -205.75002
  Iteration 3:   log likelihood = -205.03588
  Iteration 4:   log likelihood = -205.03418
  Iteration 5:   log likelihood = -205.03418

  Logistic regression                                     Number of obs =    669
                                                          LR chi2(14)   = 281.59
                                                          Prob > chi2   = 0.0000
  Log likelihood = -205.03418                             Pseudo R2     = 0.4071

------------------------------------------------------------------------------
        died | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
1.race_white~r |   .6674574   .2889324     2.31   0.021     .1011603    1.233755
    platelet |  -.0018765   .0017351    -1.08   0.279    -.0052772    .0015242
     birthwt |  -.0015043   .0008374    -1.80   0.072    -.0031455    .0001369
     gestage |  -.5211553   .1042435    -5.00   0.000    -.7254688   -.3168418
   2.placedev |   .2587623   .3099849     0.83   0.404     -.348797    .8663216
   1.multpreg |   .4112682   .3440374     1.20   0.232    -.2630326    1.085569
    1.betamet |  -.2702313   .2768959    -0.98   0.329    -.8129374    .2724747
    1.tocolys |  -.1623952   .3172102    -0.51   0.609    -.7841158    .4593254
    2.modedev |  -.0363512   .2778797    -0.13   0.896    -.5809855     .508283
       apgar1 |  -.0326145   .0570096    -0.57   0.567    -.1443511    .0791222
     1.ventil |   .6936039   .3660787     1.89   0.058    -.0238972    1.411105
     1.pneumo |   1.826412   .3078401     5.93   0.000     1.223057    2.429768
        1.pda |  -.3204829   .2878455    -1.11   0.266    -.8846497    .2436838
    2.sexbaby |  -.0524902   .2626692    -0.20   0.842    -.5673124    .4623319
        _cons |   13.90303   2.473213     5.62   0.000     9.055622    18.75044
------------------------------------------------------------------------------

132 estimates store bwt_continuous

133
134 * Model with birthweight as categorical
135 logit died i.race_white_other platelet i.birthwt_cat gestage i.placedev i.multpreg i
  > .betamet i.tocolys i.modedev apgar1 i.ventil i.pneumo i.pda i.sexbaby

  Iteration 0:   log likelihood =  -345.8274
  Iteration 1:   log likelihood = -228.48336
  Iteration 2:   log likelihood = -207.41818
  Iteration 3:   log likelihood =  -206.6295
  Iteration 4:   log likelihood = -206.62789
  Iteration 5:   log likelihood = -206.62789
```

```
Logistic regression                                     Number of obs =    669
                                                        LR chi2(14)   = 278.40
                                                        Prob > chi2   = 0.0000
Log likelihood = -206.62789                             Pseudo R2     = 0.4025
```

died	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
1.race_white_other	.704464	.2879205	2.45	0.014	.1401501	1.268778
platelet	-.0023569	.0017199	-1.37	0.171	-.0057279	.001014
1.birthwt_cat	-.0820385	.3299519	-0.25	0.804	-.7287324	.5646554
gestage	-.6501662	.0897014	-7.25	0.000	-.8259778	-.4743546
2.placedev	.2700039	.30717	0.88	0.379	-.3320383	.872046
1.multpreg	.4315157	.3409376	1.27	0.206	-.2367097	1.099741
1.betamet	-.2623863	.2751302	-0.95	0.340	-.8016317	.276859
1.tocolys	-.1875829	.3144341	-0.60	0.551	-.8038625	.4286967
2.mododev	-.0611908	.2769357	-0.22	0.825	-.6039748	.4815932
apgar1	-.0440548	.0565921	-0.78	0.436	-.1549734	.0668638
1.ventil	.7244835	.3668896	1.97	0.048	.0053931	1.443574
1.pneumo	1.741798	.3041059	5.73	0.000	1.145762	2.337835
1.pda	-.3144338	.2867846	-1.10	0.273	-.8765214	.2476538
2.sexbaby	-.1021423	.2611279	-0.39	0.696	-.6139436	.409659
_cons	16.1815	2.463262	6.57	0.000	11.3536	21.00941

```
136 estimates store bwt_categorical

137
138 * Compare the two models
139 estimates stats bwt_continuous bwt_categorical
```

Akaike's information criterion and Bayesian information criterion

Model	N	ll(null)	ll(model)	df	AIC	BIC
bwt_contin~s	669	-345.8274	-205.0342	15	440.0684	507.6551
bwt_catego~l	669	-345.8274	-206.6279	15	443.2558	510.8425

Note: BIC uses N = number of observations. See **[R] BIC note**.

```
140
141 * Both AIC and BIC agreed that birthweight is better analyzed as a continuous
142 * (numerical) variable. Birthweight will be kept as continuous going forward.
143
144 *CHECK FOR COLLINEARITY
145 * VIF cannot be run directly after logistic regression. We first predict the
146 * logit (the linear predictor), then run a linear regression using the logit
147 * as the outcome, and compute VIF from that linear model.
148
149 * Fit the full logistic model with finalised variable codings
150 logit died i.race_white_other platelet birthwt gestage i.placedev i.multpreg i.betam
  > et i.tocolys i.mododev apgar1 i.ventil i.pneumo i.pda i.sexbaby

Iteration 0:   log likelihood =  -345.8274
Iteration 1:   log likelihood = -225.79538
Iteration 2:   log likelihood = -205.75002
Iteration 3:   log likelihood = -205.03588
Iteration 4:   log likelihood = -205.03418
Iteration 5:   log likelihood = -205.03418

Logistic regression                                     Number of obs =    669
                                                        LR chi2(14)   = 281.59
                                                        Prob > chi2   = 0.0000
Log likelihood = -205.03418                             Pseudo R2     = 0.4071
```

died	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
1.race_white_other	.6674574	.2889324	2.31	0.021	.1011603	1.233755
platelet	-.0018765	.0017351	-1.08	0.279	-.0052772	.0015242
birthwt	-.0015043	.0008374	-1.80	0.072	-.0031455	.0001369
gestage	-.5211553	.1042435	-5.00	0.000	-.7254688	-.3168418
2.placedev	.2587623	.3099849	0.83	0.404	-.348797	.8663216
1.multpreg	.4112682	.3440374	1.20	0.232	-.2630326	1.085569
1.betamet	-.2702313	.2768959	-0.98	0.329	-.8129374	.2724747
1.tocolys	-.1623952	.3172102	-0.51	0.609	-.7841158	.4593254
2.modedev	-.0363512	.2778797	-0.13	0.896	-.5809855	.508283
apgar1	-.0326145	.0570096	-0.57	0.567	-.1443511	.0791222
1.ventil	.6936039	.3660787	1.89	0.058	-.0238972	1.411105
1.pneumo	1.826412	.3078401	5.93	0.000	1.223057	2.429768
1.pda	-.3204829	.2878455	-1.11	0.266	-.8846497	.2436838
2.sexbaby	-.0524902	.2626692	-0.20	0.842	-.5673124	.4623319
_cons	13.90303	2.473213	5.62	0.000	9.055622	18.75044

```
151
152 * Predict the logit (linear prediction)
153 predict Logit, xb

154
155 * Run linear regression with logit as outcome to allow VIF computation
156 regress Logit race_white_other platelet birthwt gestage i.placedev i.multpreg i.beta
  > met i.tocolys i.modedev apgar1 i.ventil i.pneumo i.pda i.sexbaby
```

Source	SS	df	MS		
Model	3285.78267	14	234.698762	Number of obs	= 669
				F(14, 654)	= .
Residual	0	654	0	Prob > F	= .
				R-squared	= 1.0000
				Adj R-squared	= 1.0000
Total	3285.78267	668	4.91883633	Root MSE	= 0

Logit	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
race_white_other	.6674574	.	.	.	.	.
platelet	-.0018765	.	.	.	.	.
birthwt	-.0015043	.	.	.	.	.
gestage	-.5211553	.	.	.	.	.
2.placedev	.2587623	.	.	.	.	.
1.multpreg	.4112682	.	.	.	.	.
1.betamet	-.2702313	.	.	.	.	.
1.tocolys	-.1623952	.	.	.	.	.
2.modedev	-.0363512	.	.	.	.	.
apgar1	-.0326145	.	.	.	.	.
1.ventil	.6936039	.	.	.	.	.
1.pneumo	1.826412	.	.	.	.	.
1.pda	-.3204829	.	.	.	.	.
2.sexbaby	-.0524902	.	.	.	.	.
_cons	13.90303	.	.	.	.	.

```
157
158 * Compute VIF - values >10 indicate collinearity
159 estat vif
```

Variable	VIF	1/VIF
race~e_other	1.13	0.886958
platelet	1.25	0.798083
birthwt	2.64	0.378865
gestage	2.66	0.376266
2.placedev	1.11	0.898551
1.multpreg	1.09	0.918641
1.betamet	1.18	0.846617
1.tocolys	1.13	0.886703
2.modedev	1.15	0.872069
apgar1	1.31	0.764492

1.ventil	**1.63** **0.614848**
1.pneumo	**1.23** **0.812403**
1.pda	**1.18** **0.847686**
2.sexbaby	**1.05** **0.950968**
Mean VIF	**1.41**

```
160
161 * No VIF exceeded 10 - no multicollinearity detected between predictors.
162
163 * STEP 2: CHECK FOR OUTLIERS
164 * We assess overall influence using Dbeta (cutoff: max < 1) and
165 * leverage (cutoff: max < 0.5) after the logistic model.
166
167 * Re-fit the logistic model
168 logit died i.race_white_other platelet birthwt gestage i.placedev i.multpreg i.betam
  > et i.tocolys i.mododev apgar1 i.ventil i.pneumo i.pda i.sexbaby

  Iteration 0:   log likelihood =  -345.8274
  Iteration 1:   log likelihood = -225.79538
  Iteration 2:   log likelihood = -205.75002
  Iteration 3:   log likelihood = -205.03588
  Iteration 4:   log likelihood = -205.03418
  Iteration 5:   log likelihood = -205.03418

  Logistic regression                                     Number of obs =    669
                                                          LR chi2(14)   = 281.59
                                                          Prob > chi2   = 0.0000
  Log likelihood = -205.03418                             Pseudo R2     = 0.4071
```

died	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
1.race_white_other	.6674574	.2889324	2.31	0.021	.1011603	1.233755
platelet	-.0018765	.0017351	-1.08	0.279	-.0052772	.0015242
birthwt	-.0015043	.0008374	-1.80	0.072	-.0031455	.0001369
gestage	-.5211553	.1042435	-5.00	0.000	-.7254688	-.3168418
2.placedev	.2587623	.3099849	0.83	0.404	-.348797	.8663216
1.multpreg	.4112682	.3440374	1.20	0.232	-.2630326	1.085569
1.betamet	-.2702313	.2768959	-0.98	0.329	-.8129374	.2724747
1.tocolys	-.1623952	.3172102	-0.51	0.609	-.7841158	.4593254
2.mododev	-.0363512	.2778797	-0.13	0.896	-.5809855	.508283
apgar1	-.0326145	.0570096	-0.57	0.567	-.1443511	.0791222
1.ventil	.6936039	.3660787	1.89	0.058	-.0238972	1.411105
1.pneumo	1.826412	.3078401	5.93	0.000	1.223057	2.429768
1.pda	-.3204829	.2878455	-1.11	0.266	-.8846497	.2436838
2.sexbaby	-.0524902	.2626692	-0.20	0.842	-.5673124	.4623319
_cons	13.90303	2.473213	5.62	0.000	9.055622	18.75044

```
169
170 * Overall influence (Dbeta) - max should not exceed 1
171 predict dbeta, dbeta

172
173 * Leverage - values should not exceed 0.5
174 predict leverage, hat

175
176 summarize dbeta leverage
```

Variable	Obs	Mean	Std. dev.	Min	Max
dbeta	**669**	**.0244991**	**.0565576**	**4.98e-08**	**.3706556**
leverage	**669**	**.0224215**	**.0199108**	**.0002422**	**.1033039**

```
177
178 * No outliers detected - Dbeta max < 1 and leverage max < 0.5.
179
180 *ASSESS FOR INTERACTION
181 * Interaction is assessed BEFORE variable selection using backward LR.
182 * A chunk test is performed by comparing a model with interaction terms
183 * (full model) against a model without them (reduced model) using the
184 * likelihood ratio test. A non-significant result means no interaction.
185
186 * Reduced model - basic variables only, no interaction terms
187 logit died i.race_white_other platelet birthwt gestage i.placedev i.multpreg ///
  > i.betamet i.tocolys i.mododev apgar1 i.ventil i.pneumo i.pda i.sexbaby

  Iteration 0:   log likelihood =  -345.8274
  Iteration 1:   log likelihood = -225.79538
  Iteration 2:   log likelihood = -205.75002
  Iteration 3:   log likelihood = -205.03588
  Iteration 4:   log likelihood = -205.03418
  Iteration 5:   log likelihood = -205.03418

  Logistic regression                                     Number of obs =    669
                                                          LR chi2(14)   = 281.59
                                                          Prob > chi2   = 0.0000
  Log likelihood = -205.03418                             Pseudo R2     = 0.4071

------------------------------------------------------------------------------
        died | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
1.race_white_other |   .6674574   .2889324     2.31   0.021     .1011603    1.233755
    platelet |  -.0018765   .0017351    -1.08   0.279    -.0052772    .0015242
     birthwt |  -.0015043   .0008374    -1.80   0.072    -.0031455    .0001369
     gestage |  -.5211553   .1042435    -5.00   0.000    -.7254688   -.3168418
  2.placedev |   .2587623   .3099849     0.83   0.404     -.348797    .8663216
  1.multpreg |   .4112682   .3440374     1.20   0.232    -.2630326    1.085569
   1.betamet |  -.2702313   .2768959    -0.98   0.329    -.8129374    .2724747
   1.tocolys |  -.1623952   .3172102    -0.51   0.609    -.7841158    .4593254
   2.mododev |  -.0363512   .2778797    -0.13   0.896    -.5809855     .508283
      apgar1 |  -.0326145   .0570096    -0.57   0.567    -.1443511    .0791222
   1.ventil |   .6936039   .3660787     1.89   0.058    -.0238972    1.411105
   1.pneumo |   1.826412   .3078401     5.93   0.000     1.223057    2.429768
     1.pda |  -.3204829   .2878455    -1.11   0.266    -.8846497    .2436838
   2.sexbaby |  -.0524902   .2626692    -0.20   0.842    -.5673124    .4623319
       _cons |   13.90303   2.473213     5.62   0.000     9.055622    18.75044
------------------------------------------------------------------------------

188 estimates store reducedmodel

189
190 * Full model - basic variables plus plausible interaction terms
191 * Interaction terms are formed between the key predictors identified
192 * from clinical relevance and preliminary inspection
193 logit died i.race_white_other platelet birthwt gestage i.placedev i.multpreg ///
  > i.betamet i.tocolys i.mododev apgar1 i.ventil i.pneumo i.pda i.sexbaby ///
  > i.pneumo#c.birthwt i.pneumo#c.gestage i.pneumo#i.race_white_other ///
  > i.pneumo#i.ventil c.birthwt#c.gestage c.birthwt#i.race_white_other ///
  > c.birthwt#i.ventil c.gestage#i.race_white_other ///
  > c.gestage#i.ventil i.race_white_other#i.ventil

  Iteration 0:   log likelihood =  -345.8274
  Iteration 1:   log likelihood = -216.37386
  Iteration 2:   log likelihood = -201.68249
  Iteration 3:   log likelihood = -198.89865
  Iteration 4:   log likelihood = -198.73478
  Iteration 5:   log likelihood = -198.73416
  Iteration 6:   log likelihood = -198.73416

  Logistic regression                                     Number of obs =    669
                                                          LR chi2(24)   = 294.19
                                                          Prob > chi2   = 0.0000
  Log likelihood = -198.73416                             Pseudo R2     = 0.4253
```

died	Coefficient	Std. err.	z	P>|z|	[95% conf. interval]
1.race_white_other	-2.675536	5.323294	-0.50	0.615	-13.109 7.757929
platelet	-.0019537	.0018024	-1.08	0.278	-.0054864 .001579
birthwt	-.0177921	.0080481	-2.21	0.027	-.0335662 -.020181
gestage	-1.159799	.426883	-2.72	0.007	-1.996475 -.323124
2.placedev	.2532356	.3162476	0.80	0.423	-.3665983 .730696
1.multpreg	.3176437	.3545994	0.90	0.370	-.3773584 1.012646
1.betamet	-.2351744	.2902463	-0.81	0.418	-.8040467 .336978
1.tocolys	-.1525932	.3256962	-0.47	0.639	-.790946 .857597
2.modedev	-.0523102	.2871555	-0.18	0.855	-.6151246 .505043
apgar1	-.0361882	.0591425	-0.61	0.541	-.1521054 .097289
1.ventil	-2.615732	6.309091	-0.41	0.678	-14.98132 9.749859
1.pneumo	1.24098	5.327937	0.23	0.816	-9.201585 11.68354
1.pda	-.2637997	.2871886	-0.92	0.358	-.826679 .299797
2.sexbaby	.0208885	.2701271	0.08	0.938	-.5085509 .550279
pneumo#c.birthwt 1	-.0013291	.001783	-0.75	0.456	-.0048236 .021654
pneumo#c.gestage 1	.0777122	.2229527	0.35	0.727	-.3592672 .514915
pneumo#race_white_other 1 1	.3880186	.6154502	0.63	0.528	-.8182417 1.594279
pneumo#ventil 1 1	-.5546323	1.511786	-0.37	0.714	-3.517678 2.408414
c.birthwt#c.gestage	.0004903	.0002688	1.82	0.068	-.0000367 .000172
race_white_other#c.birthwt 1	-.0003068	.0019214	-0.16	0.873	-.0040726 .003459
ventil#c.birthwt 1	.0039759	.002649	1.50	0.133	-.001216 .009167
race_white_other#c.gestage 1	.1806288	.2398351	0.75	0.451	-.2894393 .650697
ventil#c.gestage 1	.0351893	.2929483	0.12	0.904	-.5389787 .609357
race_white_other#ventil 1 1	-1.631584	1.116162	-1.46	0.144	-3.819222 .5

```
> 560536
                    _cons |   32.79537   10.15344     3.23   0.001     12.89499    52
> .69576
------------------------------------------------------------------------------
      _____
```

```
194 estimates store fullmodel

195
196 * Likelihood ratio test comparing full model (with interaction) to reduced model
197 * A non-significant p-value means no evidence of interaction - reduced is better
198 lrtest fullmodel reducedmodel, stats

  Likelihood-ratio test
  Assumption: reducedmodel nested within fullmodel

  LR chi2(10) =  12.60
  Prob > chi2 = 0.2469

  Akaike's information criterion and Bayesian information criterion

-----------------------------------------------------------------------------
       Model |          N   ll(null)  ll(model)      df        AIC        BIC
-------------+---------------------------------------------------------------
reducedmodel |        669  -345.8274  -205.0342      15   440.0684   507.6551
   fullmodel |        669  -345.8274  -198.7342      25   447.4683   560.1129
-----------------------------------------------------------------------------
  Note: BIC uses N = number of observations. See [R] BIC note.

199
200 * The likelihood ratio test was not significant. No interaction was detected.
201 * The reduced model (without interaction terms) is preferred.
202
203 * VARIABLE SELECTION - BACKWARD LIKELIHOOD RATIO
204 * Backward LR stepwise is used to identify independent predictors of death.
205
206 stepwise, pr(0.05): logit died i.race_white_other platelet birthwt gestage i.placede
  > v i.multpreg i.betamet i.tocolys i.modedev apgar1 i.ventil i.pneumo i.pda i.sexbaby
  note: 0b.race_white_other omitted because of estimability.
  note: 1b.placedev omitted because of estimability.
  note: 0b.multpreg omitted because of estimability.
  note: 0b.betamet omitted because of estimability.
  note: 0b.tocolys omitted because of estimability.
  note: 1b.modedev omitted because of estimability.
  note: 0b.ventil omitted because of estimability.
  note: 0b.pneumo omitted because of estimability.
  note: 0b.pda omitted because of estimability.
  note: 1b.sexbaby omitted because of estimability.

  Wald test, begin with full model:
  p = 0.8959 >= 0.0500, removing 2.modedev
  p = 0.8412 >= 0.0500, removing 2.sexbaby
  p = 0.6121 >= 0.0500, removing 1.tocolys
  p = 0.5716 >= 0.0500, removing apgar1
  p = 0.4258 >= 0.0500, removing 2.placedev
  p = 0.2458 >= 0.0500, removing 1.pda
  p = 0.2537 >= 0.0500, removing platelet
  p = 0.2023 >= 0.0500, removing 1.multpreg
  p = 0.2020 >= 0.0500, removing 1.betamet

  Logistic regression                                     Number of obs =    669
                                                          LR chi2(5)    = 274.39
                                                          Prob > chi2   = 0.0000
  Log likelihood = -208.63109                             Pseudo R2     = 0.3967
```

died	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
1.race_white_other	.6996867	.2788048	2.51	0.012	.1532393	1.246134
1.pneumo	1.87839	.3008724	6.24	0.000	1.288691	2.46809
birthwt	-.0019013	.0008042	-2.36	0.018	-.0034774	-.0003251
gestage	-.4852203	.0999582	-4.85	0.000	-.6811347	-.2893058
1.ventil	.7584576	.3361824	2.26	0.024	.0995523	1.417363
_cons	12.5697	2.273295	5.53	0.000	8.11412	17.02527

```
207
208 * Independent predictors retained: race_white_other, birthwt, gestage,
209 * ventil, pneumo
210
211 * ASSESS FOR CONFOUNDING
212 * Variables dropped during stepwise are added back one at a time to check
213 * whether they confound the relationship between the retained predictors
214 * and the outcome. A change of 10% or more in any odds ratio indicates
215 * confounding - that variable is then retained in the final model.
216
217 * Base model with independent predictors from stepwise
218 logit died i.race_white_other birthwt gestage i.ventil i.pneumo, or
```

```
Iteration 0:   log likelihood =  -345.8274
Iteration 1:   log likelihood = -229.10865
Iteration 2:   log likelihood = -209.30083
Iteration 3:   log likelihood = -208.63271
Iteration 4:   log likelihood = -208.63109
Iteration 5:   log likelihood = -208.63109

Logistic regression                                     Number of obs =    669
                                                        LR chi2(5)    = 274.39
                                                        Prob > chi2   = 0.0000
Log likelihood = -208.63109                             Pseudo R2     = 0.3967
```

died	Odds ratio	Std. err.	z	P>|z|	[95% conf.	interval]
1.race_white_other	2.013122	.5612681	2.51	0.012	1.165604	3.476876
birthwt	.9981005	.0008027	-2.36	0.018	.9965286	.9996749
gestage	.6155616	.0615304	-4.85	0.000	.5060424	.7487832
1.ventil	2.134981	.717743	2.26	0.024	1.104676	4.126225
1.pneumo	6.542965	1.968598	6.24	0.000	3.628035	11.79988
_cons	287706.8	654042.5	5.53	0.000	3341.317	2.48e+07

Note: **_cons** estimates baseline odds.

```
219
220 * Add betamethasone
221 logit died i.race_white_other birthwt gestage i.ventil i.pneumo i.betamet, or
```

```
Iteration 0:   log likelihood =  -345.8274
Iteration 1:   log likelihood = -228.31503
Iteration 2:   log likelihood = -208.47918
Iteration 3:   log likelihood = -207.81368
Iteration 4:   log likelihood = -207.81206
Iteration 5:   log likelihood = -207.81206

Logistic regression                                     Number of obs =    669
                                                        LR chi2(6)    = 276.03
                                                        Prob > chi2   = 0.0000
Log likelihood = -207.81206                             Pseudo R2     = 0.3991
```

died	Odds ratio	Std. err.	z	P>|z|	[95% conf.	interval]
1.race_white_other	2.002694	.558621	2.49	0.013	1.159271	3.459747
birthwt	.9981424	.000799	-2.32	0.020	.9965776	.9997098
gestage	.6170306	.0613986	-4.85	0.000	.5076992	.7499061
1.ventil	2.097611	.7049106	2.20	0.027	1.085613	4.052981
1.pneumo	6.463182	1.943558	6.21	0.000	3.584922	11.65234
1.betamet	.7206669	.1850418	-1.28	0.202	.4356885	1.192046
_cons	301215.8	682939	5.56	0.000	3539.728	2.56e+07

Note: **_cons** estimates baseline odds.

```
222
223 * Add multiple pregnancy
224 logit died i.race_white_other birthwt gestage i.ventil i.pneumo i.multpreg, or
```

Iteration 0: log likelihood = **-345.8274**
Iteration 1: log likelihood = **-228.5188**
Iteration 2: log likelihood = **-208.68529**
Iteration 3: log likelihood = **-208.01091**
Iteration 4: log likelihood = **-208.00927**
Iteration 5: log likelihood = **-208.00927**

Logistic regression — Number of obs = **669**
LR chi2(**6**) = **275.64**
Prob > chi2 = **0.0000**
Log likelihood = **-208.00927** — Pseudo R2 = **0.3985**

died	Odds ratio	Std. err.	z	P>|z|	[95% conf.	interval]
1.race_white_other	2.01249	.5616393	2.51	0.012	1.164617	3.477638
birthwt	.9981296	.00081	-2.31	0.021	.9965433	.9997185
gestage	.6096811	.0619109	-4.87	0.000	.4996505	.743942
1.ventil	2.181061	.7378159	2.31	0.021	1.123882	4.232674
1.pneumo	6.306382	1.909164	6.08	0.000	3.484108	11.41482
1.multpreg	1.442259	.4709163	1.12	0.262	.7605273	2.73509
_cons	339089.1	780942.5	5.53	0.000	3714.886	3.10e+07

Note: **_cons** estimates baseline odds.

```
225
226 * Add platelet count
227 logit died i.race_white_other birthwt gestage i.ventil i.pneumo platelet, or
```

Iteration 0: log likelihood = **-345.8274**
Iteration 1: log likelihood = **-228.27703**
Iteration 2: log likelihood = **-208.35178**
Iteration 3: log likelihood = **-207.69069**
Iteration 4: log likelihood = **-207.68911**
Iteration 5: log likelihood = **-207.68911**

Logistic regression — Number of obs = **669**
LR chi2(**6**) = **276.28**
Prob > chi2 = **0.0000**
Log likelihood = **-207.68911** — Pseudo R2 = **0.3994**

died	Odds ratio	Std. err.	z	P>|z|	[95% conf.	interval]
1.race_white_other	2.009365	.5607259	2.50	0.012	1.162855	3.472098
birthwt	.9983361	.0008181	-2.03	0.042	.996734	.9999408
gestage	.6046581	.0608301	-5.00	0.000	.4964517	.7364493
1.ventil	1.960421	.6693261	1.97	0.049	1.003997	3.82795
1.pneumo	6.398055	1.927694	6.16	0.000	3.544755	11.54808
platelet	.9977446	.0016509	-1.36	0.172	.9945142	1.000986
_cons	615523.3	1442461	5.69	0.000	6229.863	6.08e+07

Note: **_cons** estimates baseline odds.

```
228
229 * Add PDA
230 logit died i.race_white_other birthwt gestage i.ventil i.pneumo i.pda, or

  Iteration 0:   log likelihood =  -345.8274
  Iteration 1:   log likelihood = -228.67997
  Iteration 2:   log likelihood = -208.84426
  Iteration 3:   log likelihood = -208.16973
  Iteration 4:   log likelihood = -208.16811
  Iteration 5:   log likelihood = -208.16811

  Logistic regression                                     Number of obs =    669
                                                          LR chi2(6)    = 275.32
                                                          Prob > chi2   = 0.0000
  Log likelihood = -208.16811                             Pseudo R2     = 0.3981

------------------------------------------------------------------------------
        died | Odds ratio   Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
1.race_white~r |   2.02167   .5647637     2.52   0.012     1.169292    3.495407
     birthwt |   .9980948   .0008018    -2.37   0.018     .9965245    .9996675
     gestage |   .6136821   .0613892    -4.88   0.000     .5044222    .7466082
    1.ventil |   2.273286   .7808559     2.39   0.017     1.159506    4.456923
    1.pneumo |   6.757345   2.049167     6.30   0.000     3.729486    12.24343
       1.pda |   .7629507   .2159031    -0.96   0.339     .4381469    1.328536
       _cons |   319650.2     727984     5.57   0.000     3682.265    2.77e+07
------------------------------------------------------------------------------
  Note: _cons estimates baseline odds.

231
232 * Add place of delivery
233 logit died i.race_white_other birthwt gestage i.ventil i.pneumo i.placedev, or

  Iteration 0:   log likelihood =  -345.8274
  Iteration 1:   log likelihood = -228.73804
  Iteration 2:   log likelihood = -208.87885
  Iteration 3:   log likelihood = -208.20921
  Iteration 4:   log likelihood = -208.20759
  Iteration 5:   log likelihood = -208.20759

  Logistic regression                                     Number of obs =    669
                                                          LR chi2(6)    = 275.24
                                                          Prob > chi2   = 0.0000
  Log likelihood = -208.20759                             Pseudo R2     = 0.3979

------------------------------------------------------------------------------
        died | Odds ratio   Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
1.race_white~r |   2.016687   .5641631     2.51   0.012     1.165512    3.489474
     birthwt |   .9981106   .0008028    -2.35   0.019     .9965384    .9996853
     gestage |   .6154959   .0614754    -4.86   0.000     .5060665    .7485877
    1.ventil |    2.05442   .6971441     2.12   0.034     1.056437    3.995166
    1.pneumo |   6.529756   1.965925     6.23   0.000     3.619298    11.78066
  2.placedev |   1.311098   .3843075     0.92   0.355     .7381282    2.328833
       _cons |   276530.3   628203.6     5.52   0.000     3221.356    2.37e+07
------------------------------------------------------------------------------
  Note: _cons estimates baseline odds.

234
235 * Add Apgar score
```

```
236 logit died i.race_white_other birthwt gestage i.ventil i.pneumo apgar1, or

  Iteration 0:   log likelihood =  -345.8274
  Iteration 1:   log likelihood = -228.54064
  Iteration 2:   log likelihood = -208.74607
  Iteration 3:   log likelihood = -208.07966
  Iteration 4:   log likelihood = -208.07804
  Iteration 5:   log likelihood = -208.07804

  Logistic regression                                     Number of obs =    669
                                                          LR chi2(6)    = 275.50
                                                          Prob > chi2   = 0.0000
  Log likelihood = -208.07804                             Pseudo R2     = 0.3983

------------------------------------------------------------------------------
        died | Odds ratio   Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
1.race_white~r |   1.934652   .5441672     2.35   0.019     1.114762    3.357559
     birthwt |   .9982388   .0008102    -2.17   0.030     .9966521    .9998281
     gestage |   .6145374   .0612659    -4.88   0.000     .5054618    .7471509
    1.ventil |   1.960006   .6772579     1.95   0.051     .9957138    3.858162
    1.pneumo |   6.539197   1.971293     6.23   0.000     3.621787    11.80663
      apgar1 |   .9454318   .0504876    -1.05   0.293     .8514804     1.04975
       _cons |   367599.3   838511.8     5.62   0.000     4204.773    3.21e+07
------------------------------------------------------------------------------
  Note: _cons estimates baseline odds.

237
238 * Add tocolysis
239 logit died i.race_white_other birthwt gestage i.ventil i.pneumo i.tocolys, or

  Iteration 0:   log likelihood =  -345.8274
  Iteration 1:   log likelihood = -228.93642
  Iteration 2:   log likelihood = -209.11026
  Iteration 3:   log likelihood = -208.43963
  Iteration 4:   log likelihood = -208.43802
  Iteration 5:   log likelihood = -208.43802

  Logistic regression                                     Number of obs =    669
                                                          LR chi2(6)    = 274.78
                                                          Prob > chi2   = 0.0000
  Log likelihood = -208.43802                             Pseudo R2     = 0.3973

------------------------------------------------------------------------------
        died | Odds ratio   Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
1.race_white~r |   1.974422   .5534326     2.43   0.015     1.139849    3.420052
     birthwt |   .9981232   .0008025    -2.34   0.019     .9965516    .9996973
     gestage |    .613148   .0613628    -4.89   0.000     .5039396    .7460229
    1.ventil |   2.119491   .7143088     2.23   0.026     1.094865    4.103011
    1.pneumo |   6.497625   1.954068     6.22   0.000      3.60386    11.71497
    1.tocolys |   .8305412   .2493354    -0.62   0.536     .4611299    1.495888
       _cons |   334055.4   763858.5     5.56   0.000     3779.565    2.95e+07
------------------------------------------------------------------------------
  Note: _cons estimates baseline odds.

240
241 * Add sex of baby
242 logit died i.race_white_other birthwt gestage i.ventil i.pneumo i.sexbaby, or

  Iteration 0:   log likelihood =  -345.8274
  Iteration 1:   log likelihood = -229.11261
  Iteration 2:   log likelihood = -209.30197
  Iteration 3:   log likelihood = -208.61754
  Iteration 4:   log likelihood = -208.61591
  Iteration 5:   log likelihood = -208.61591

  Logistic regression                                     Number of obs =    669
                                                          LR chi2(6)    = 274.42
                                                          Prob > chi2   = 0.0000
  Log likelihood = -208.61591                             Pseudo R2     = 0.3968
```

died	Odds ratio	Std. err.	z	P>|z|	[95% conf.	interval]
1.race_white_other	2.022224	.5662234	2.51	0.012	1.168134	3.500789
birthwt	.9981192	.0008098	-2.32	0.020	.9965334	.9997076
gestage	.614533	.0617863	-4.84	0.000	.5046192	.7483878
1.ventil	2.160107	.7409736	2.25	0.025	1.102785	4.231164
1.pneumo	6.542192	1.967957	6.24	0.000	3.628051	11.79705
2.sexbaby	.9563096	.2452503	-0.17	0.862	.5785005	1.58086
_cons	299294.8	684622.7	5.51	0.000	3380.774	2.65e+07

Note: **_cons** estimates baseline odds.

```
243
244 * Add mode of delivery
245 logit died i.race_white_other birthwt gestage i.ventil i.pneumo i.modedev, or
```

```
Iteration 0:   log likelihood =  -345.8274
Iteration 1:   log likelihood = -229.10839
Iteration 2:   log likelihood = -209.30966
Iteration 3:   log likelihood = -208.61491
Iteration 4:   log likelihood = -208.61323
Iteration 5:   log likelihood = -208.61323

Logistic regression                                     Number of obs =    669
                                                        LR chi2(6)    = 274.43
                                                        Prob > chi2   = 0.0000
Log likelihood = -208.61323                             Pseudo R2     = 0.3968
```

died	Odds ratio	Std. err.	z	P>|z|	[95% conf.	interval]
1.race_white_other	2.02726	.5703921	2.51	0.012	1.167923	3.51888
birthwt	.9981094	.0008043	-2.35	0.019	.9965343	.999687
gestage	.6133099	.0625545	-4.79	0.000	.5021827	.7490284
1.ventil	2.13153	.7172377	2.25	0.024	1.102227	4.122037
1.pneumo	6.495579	1.97029	6.17	0.000	3.584468	11.77094
2.modedev	.9513965	.2508804	-0.19	0.850	.5674169	1.595221
_cons	323543	763925.5	5.37	0.000	3163.289	3.31e+07

Note: **_cons** estimates baseline odds.

```
246
247 * None of the dropped variables changed any odds ratio by 10% or more.
248 * No confounding was detected. The final model retains only the independent
249 * predictors identified from backward LR.
250
251 * FINAL MODEL
252
253 logit died i.race_white_other birthwt gestage i.ventil i.pneumo
```

```
Iteration 0:   log likelihood =  -345.8274
Iteration 1:   log likelihood = -229.10865
Iteration 2:   log likelihood = -209.30083
Iteration 3:   log likelihood = -208.63271
Iteration 4:   log likelihood = -208.63109
Iteration 5:   log likelihood = -208.63109

Logistic regression                                     Number of obs =    669
                                                        LR chi2(5)    = 274.39
                                                        Prob > chi2   = 0.0000
Log likelihood = -208.63109                             Pseudo R2     = 0.3967
```

died	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
1.race_white_other	.6996867	.2788048	2.51	0.012	.1532393	1.246134
birthwt	-.0019013	.0008042	-2.36	0.018	-.0034774	-.0003251
gestage	-.4852203	.0999582	-4.85	0.000	-.6811347	-.2893058
1.ventil	.7584576	.3361824	2.26	0.024	.0995523	1.417363
1.pneumo	1.87839	.3008724	6.24	0.000	1.288691	2.46809
_cons	12.5697	2.273295	5.53	0.000	8.11412	17.02527

```
254 estimates store finalmodel

255
256 * Display as odds ratios with 95% confidence intervals
257 logistic died i.race_white_other birthwt gestage i.ventil i.pneumo
```

Logistic regression — Number of obs = 669; LR chi2(5) = 274.39; Prob > chi2 = 0.0000; Log likelihood = -208.63109; Pseudo R2 = 0.3967

died	Odds ratio	Std. err.	z	P>|z|	[95% conf.	interval]
1.race_white_other	2.013122	.5612681	2.51	0.012	1.165604	3.476876
birthwt	.9981005	.0008027	-2.36	0.018	.9965286	.9996749
gestage	.6155616	.0615304	-4.85	0.000	.5060424	.7487832
1.ventil	2.134981	.717743	2.26	0.024	1.104676	4.126225
1.pneumo	6.542965	1.968598	6.24	0.000	3.628035	11.79988
_cons	287706.8	654042.5	5.53	0.000	3341.317	2.48e+07

Note: **_cons** estimates baseline odds.

```
258
259 * GOODNESS OF FIT
260 * The Hosmer-Lemeshow test divides observations into 10 groups by predicted
261 * probability and compares observed to expected frequencies. A non-significant
262 * p-value (p>0.05) indicates a good fit.
263
264 logistic died i.race_white_other birthwt gestage i.ventil i.pneumo
```

Logistic regression — Number of obs = 669; LR chi2(5) = 274.39; Prob > chi2 = 0.0000; Log likelihood = -208.63109; Pseudo R2 = 0.3967

died	Odds ratio	Std. err.	z	P>|z|	[95% conf.	interval]
1.race_white_other	2.013122	.5612681	2.51	0.012	1.165604	3.476876
birthwt	.9981005	.0008027	-2.36	0.018	.9965286	.9996749
gestage	.6155616	.0615304	-4.85	0.000	.5060424	.7487832
1.ventil	2.134981	.717743	2.26	0.024	1.104676	4.126225
1.pneumo	6.542965	1.968598	6.24	0.000	3.628035	11.79988
_cons	287706.8	654042.5	5.53	0.000	3341.317	2.48e+07

Note: **_cons** estimates baseline odds.

```
265 estat gof, group(10)
  note: obs collapsed on 10 quantiles of estimated probabilities.

  Goodness-of-fit test after logistic model
  Variable: died

   Number of observations =    669
         Number of groups =     10
  Hosmer-Lemeshow chi2(8) =  10.10
              Prob > chi2 = 0.2578
```

```
266
267 * A non-significant Hosmer-Lemeshow test indicates the model fits the data well.
268
269
270
271
272 log close
      name:  <unnamed>
       log:  E:\Documents\STATA\assignment 2_Number one baby outcomes.smcl
  log type:  smcl
 closed on:  14 May 2026, 09:58:33
```